LAMPIRAN PERANCANGAN ELEKTRIK
ROBOT WELDING ARM MANIPULATOR 6-DOF

Dirancang Oleh :

Alfonsus Giovanni Mahendra Putra / Teknik Listrik Industri / 40040621650042

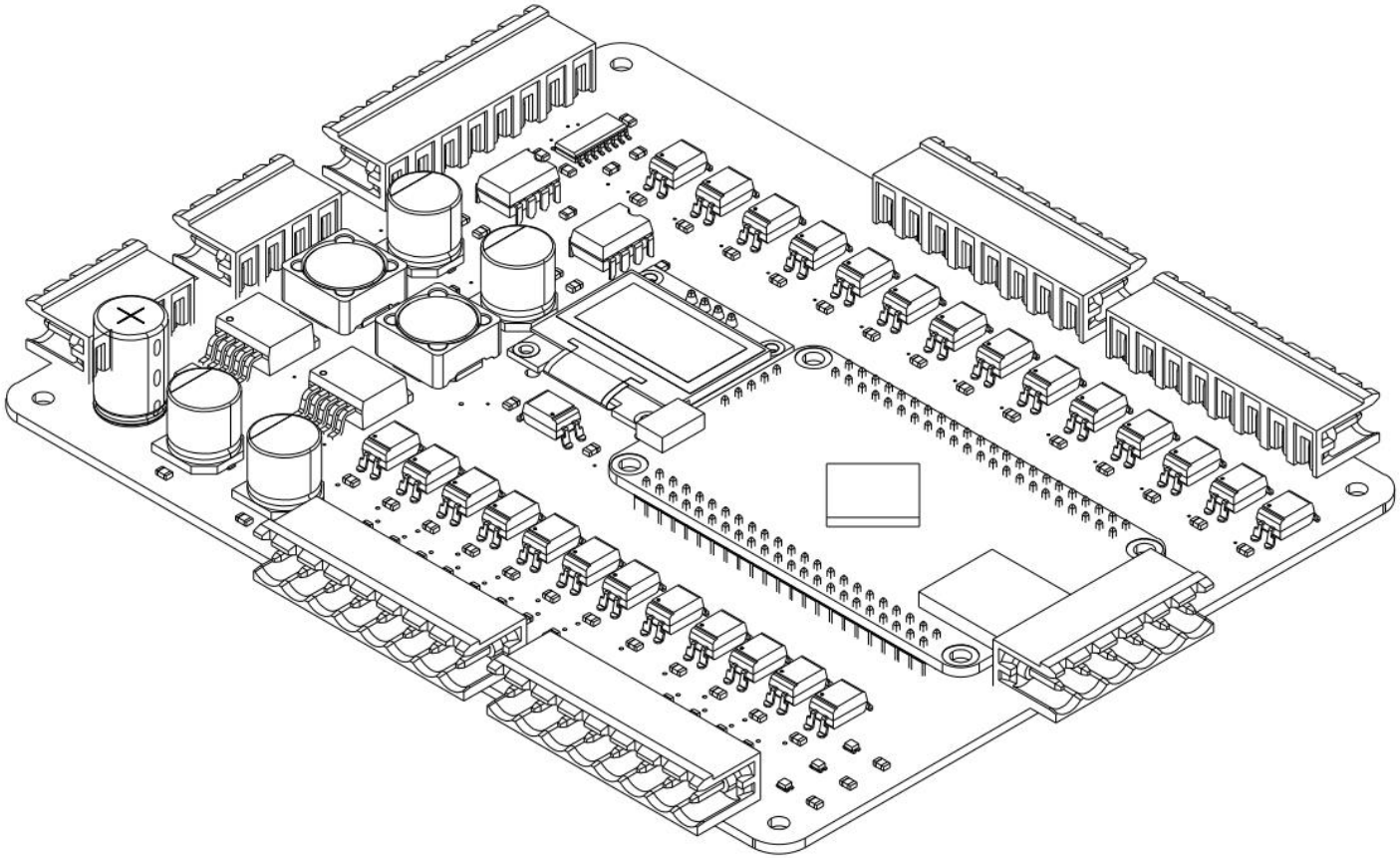
DEPARTEMEN TEKNOLOGI INDUSTRI

FAKULTAS SEKOLAH VOKASI

UNIVERSITAS DIPONEGORO

2024

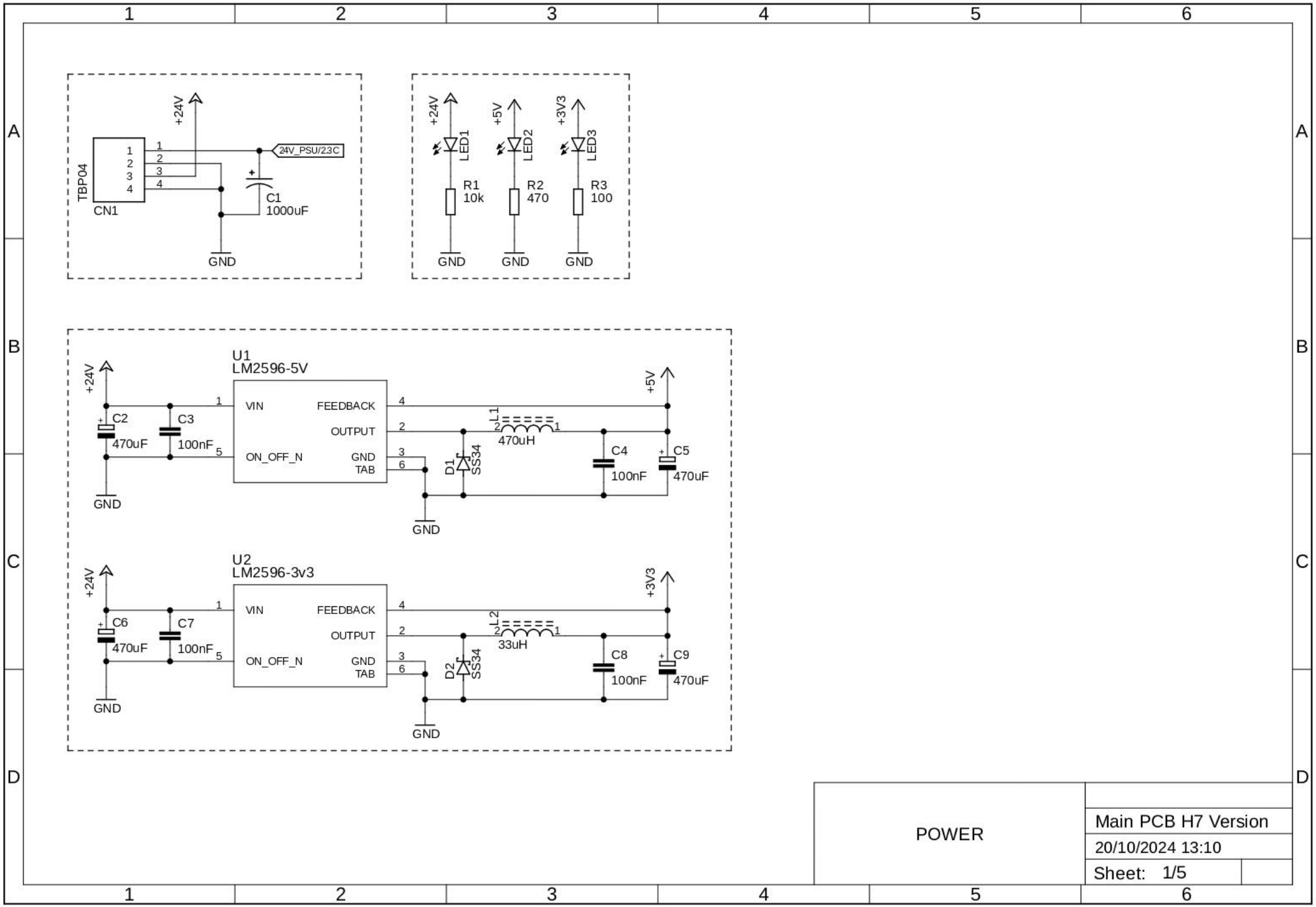
PCB KONTROLER UTAMA



Spesifikasi:

1. Microcontroller STM32H743VIT6
2. 24VDC Supply
3. 5VDC and 3,3VDC Output
4. Dual Emergency Stop
5. 18 Channels Digital Input
6. 12 Channels Digital Output
7. RS-232 Communication
8. OLED LCD Interface

SCHEMATIC DIAGRAM PCB KONTROLER UTAMA



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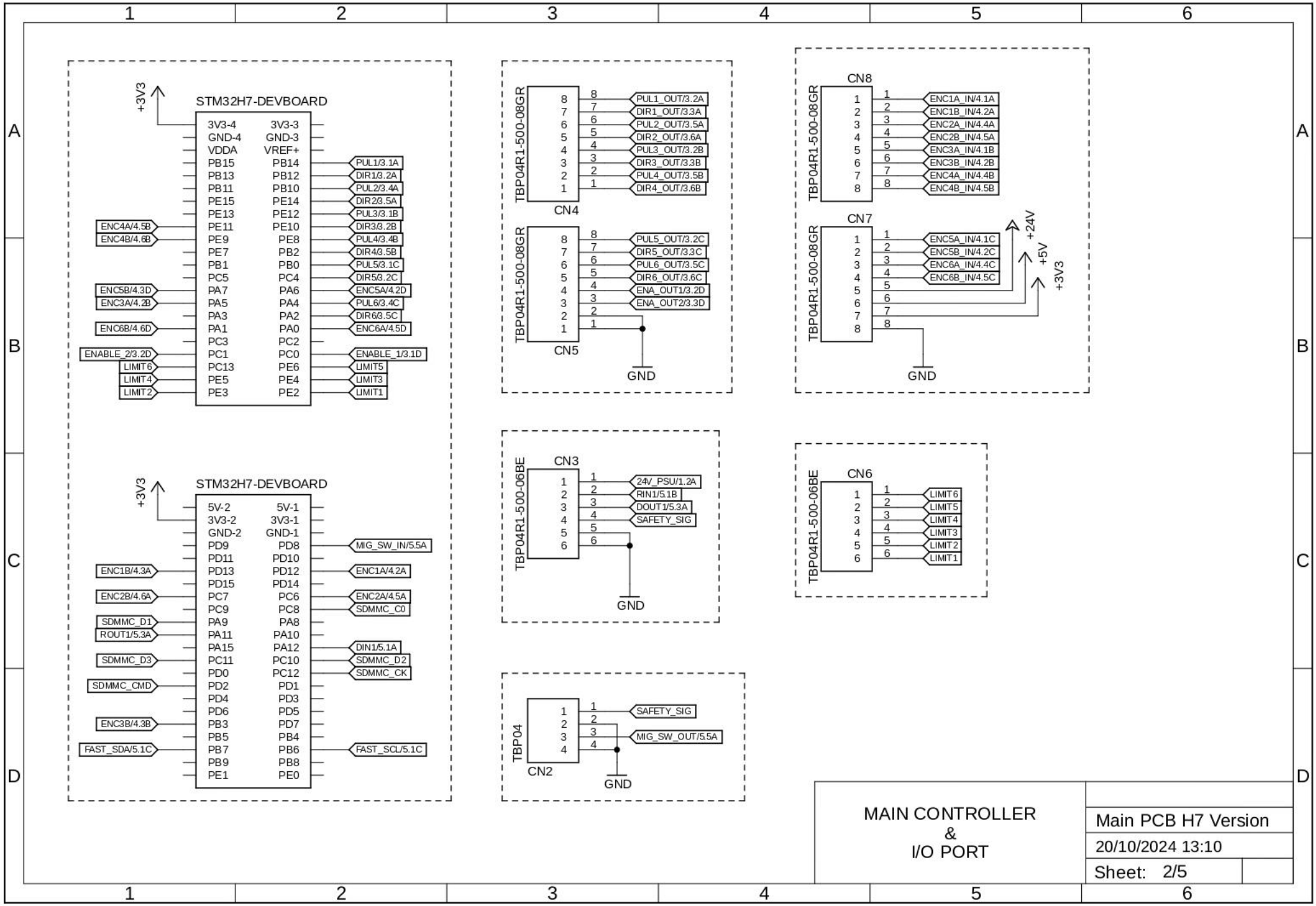

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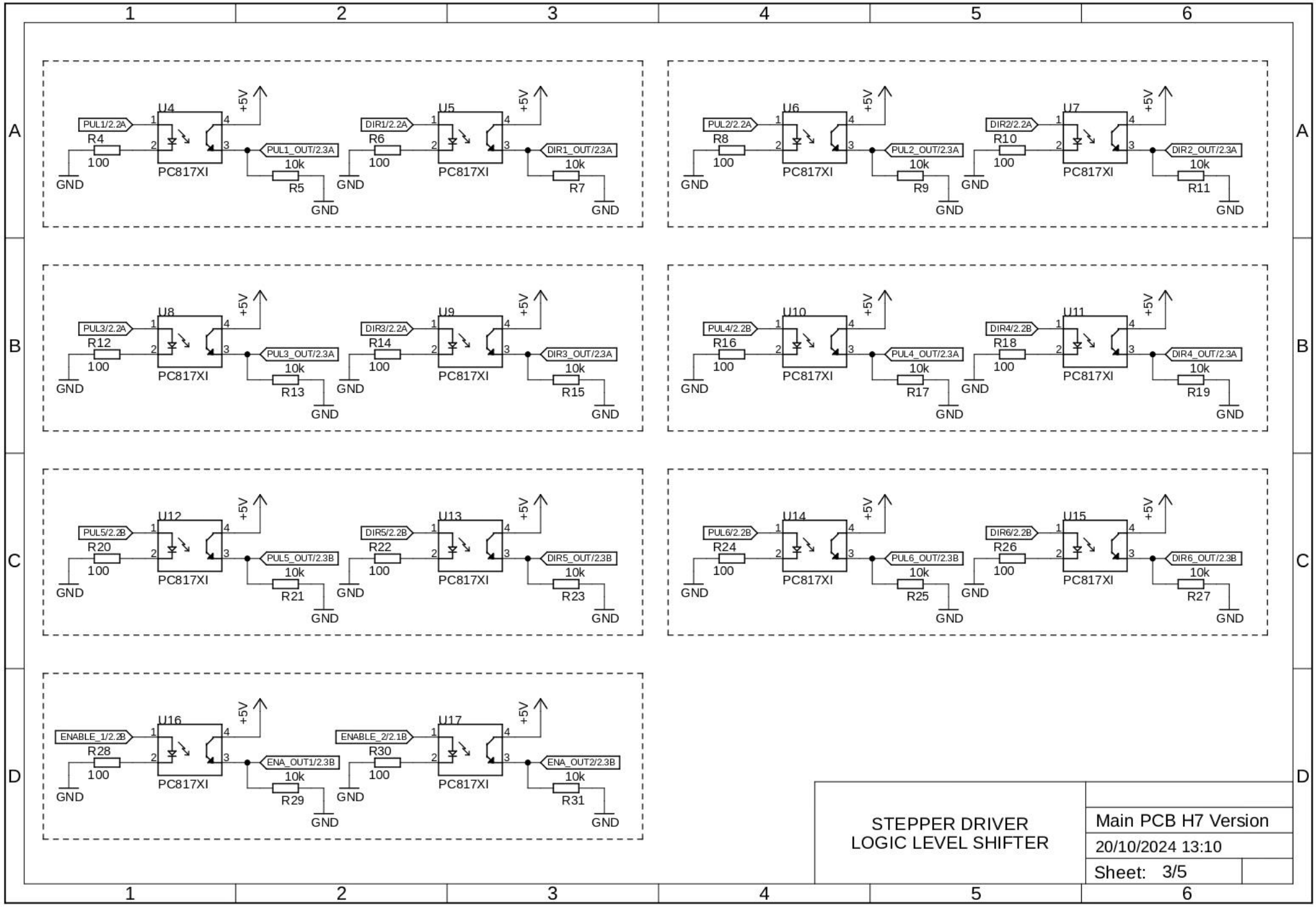

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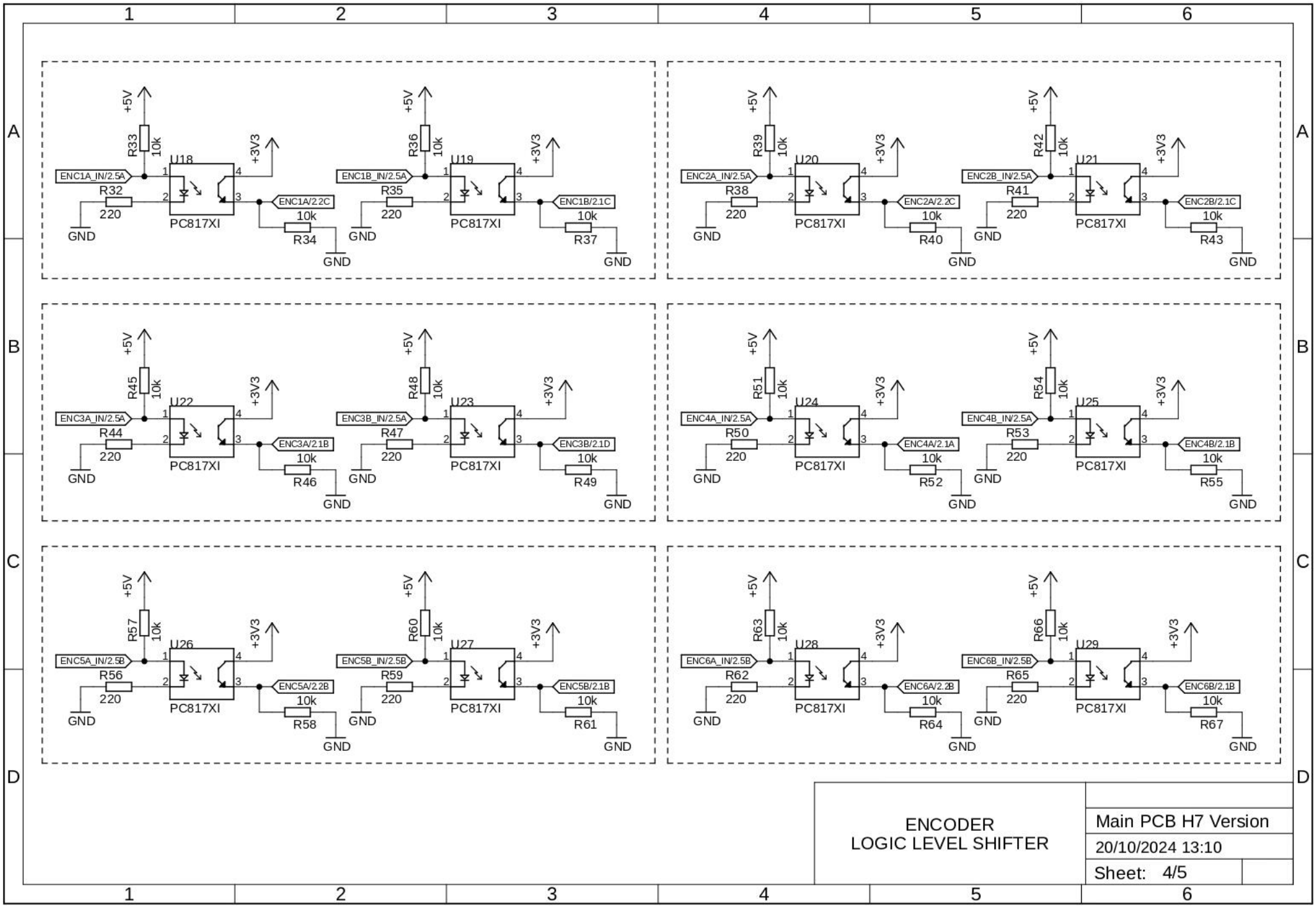
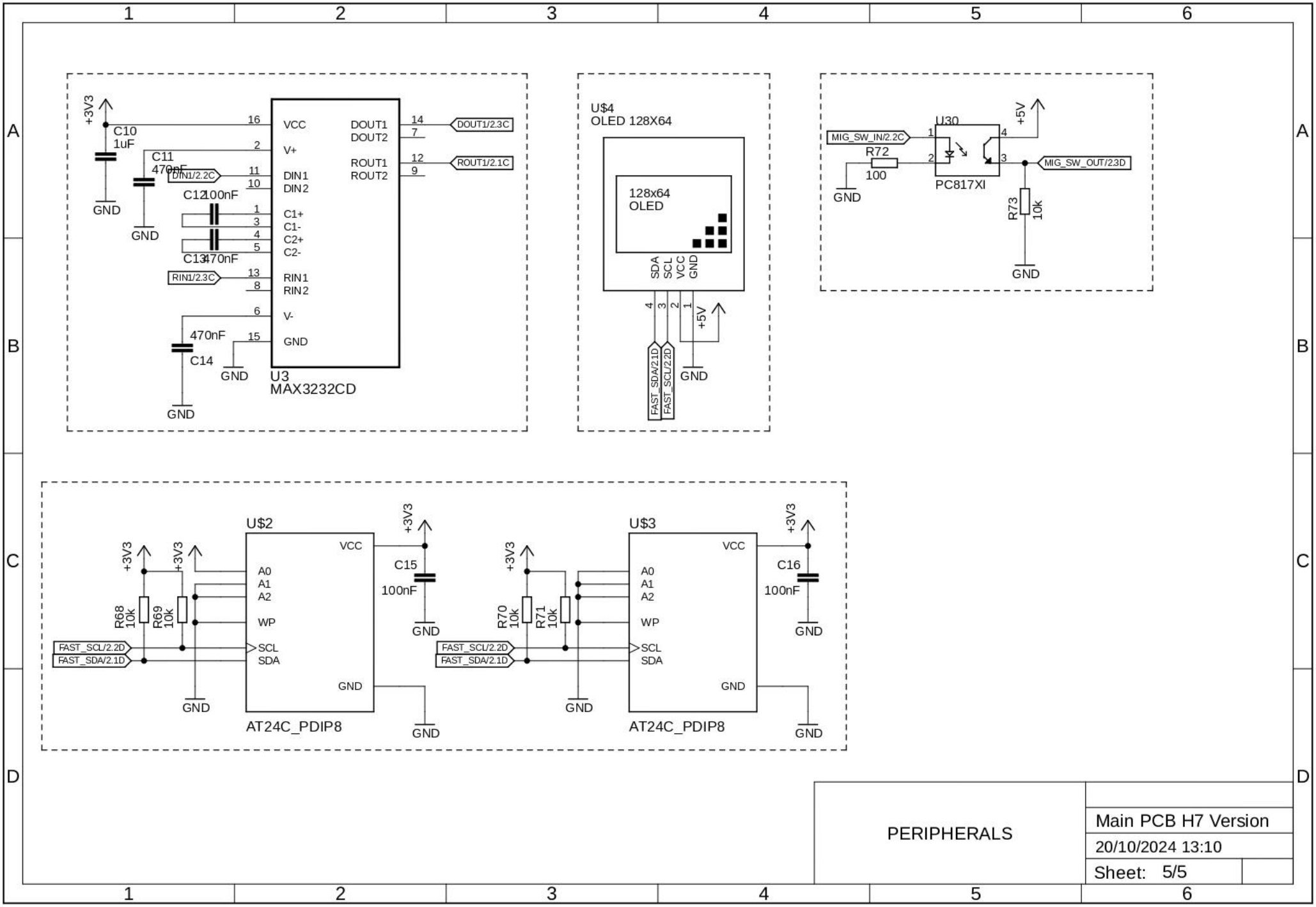
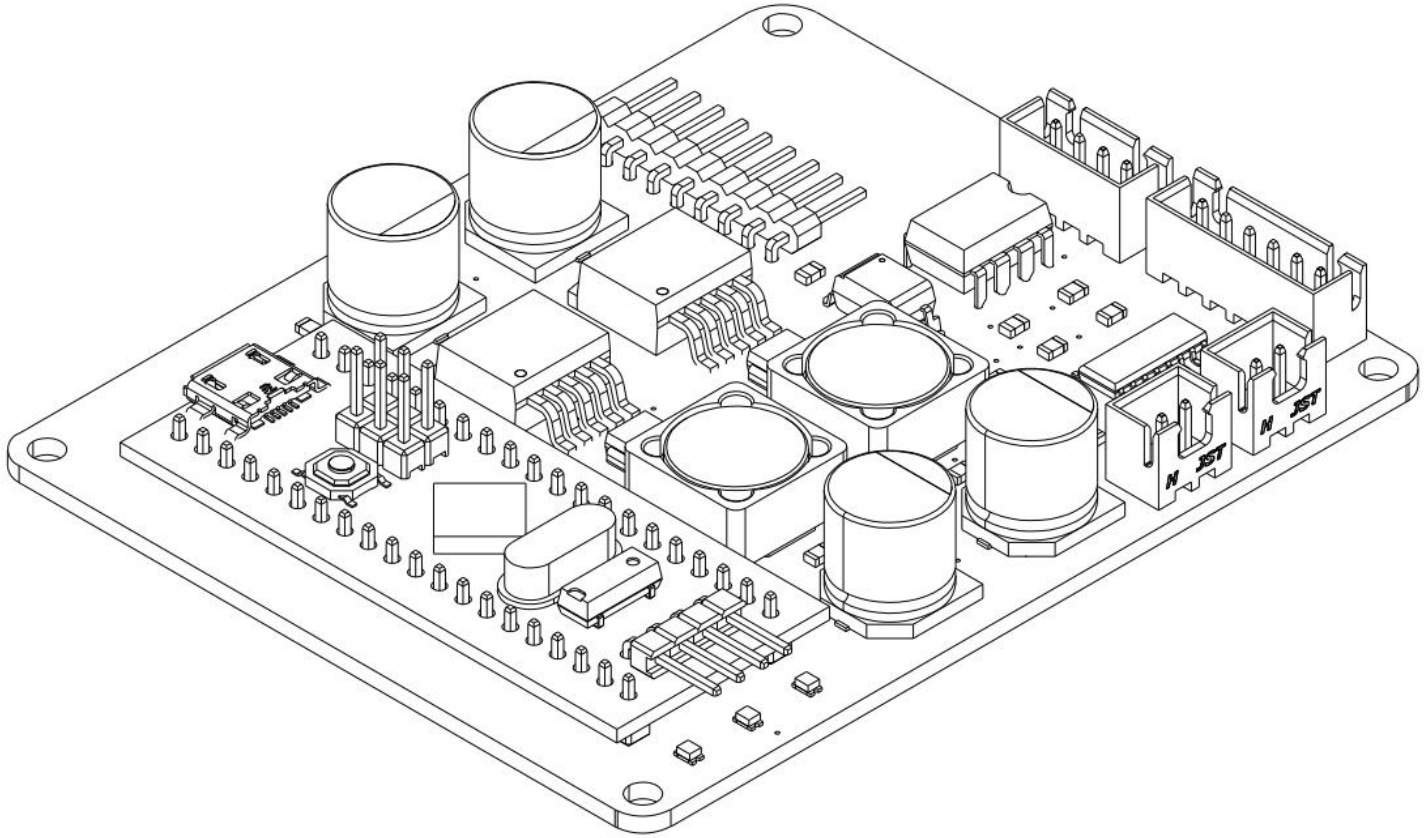


SCHEMATIC DIAGRAM PCB KONTROLER UTAMA



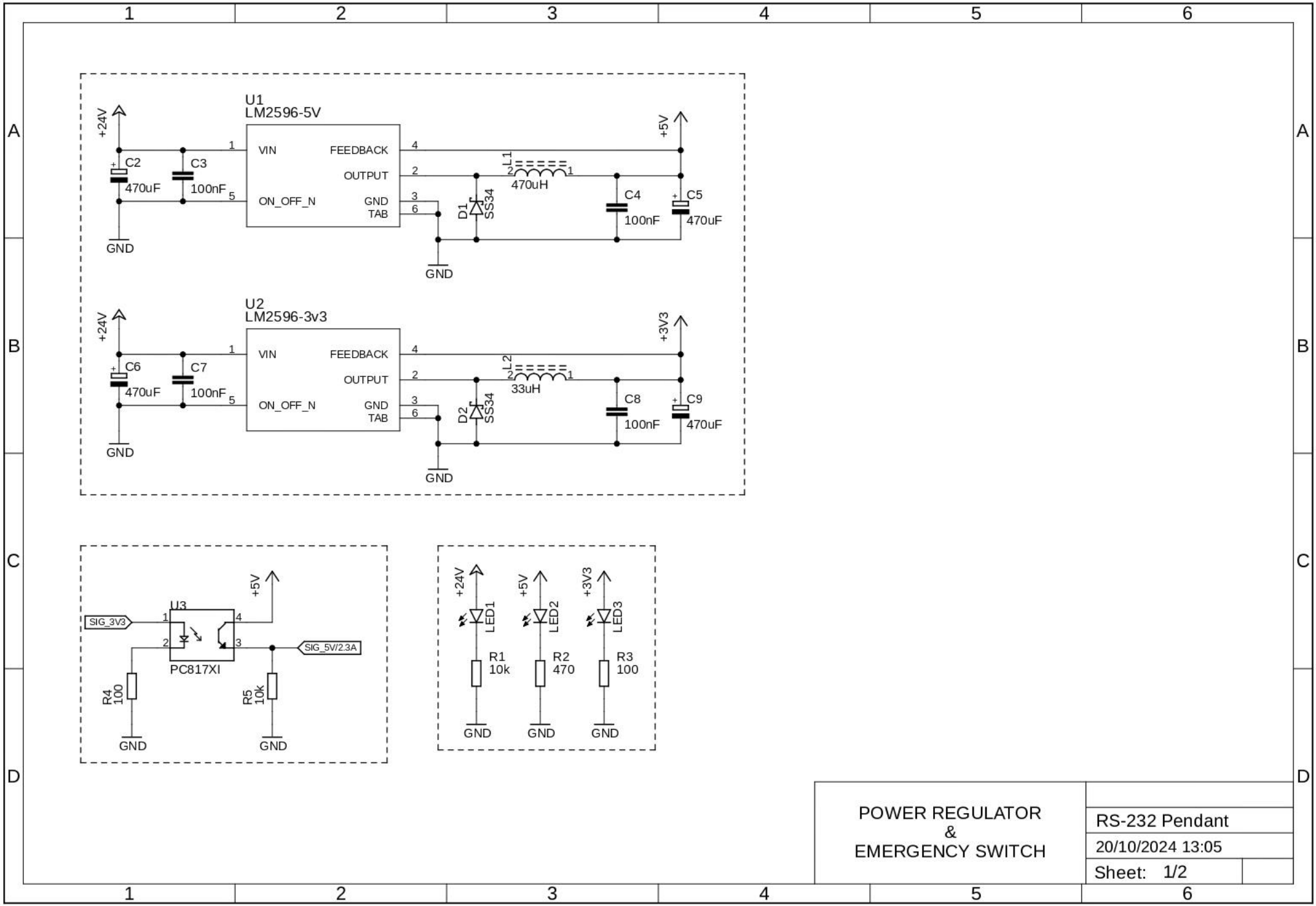
PCB TEACH PENDANT



Spesifikasi:

1. Mikrokontroler STM32F103C8T6
2. 24VDC Supply
3. On/Off Selector Switch
4. Interface LCD 20x4
5. Keypad Matrix 5x4
6. Emergency Stop

SCHEMATIC DIAGRAM PCB TEACH PENDANT



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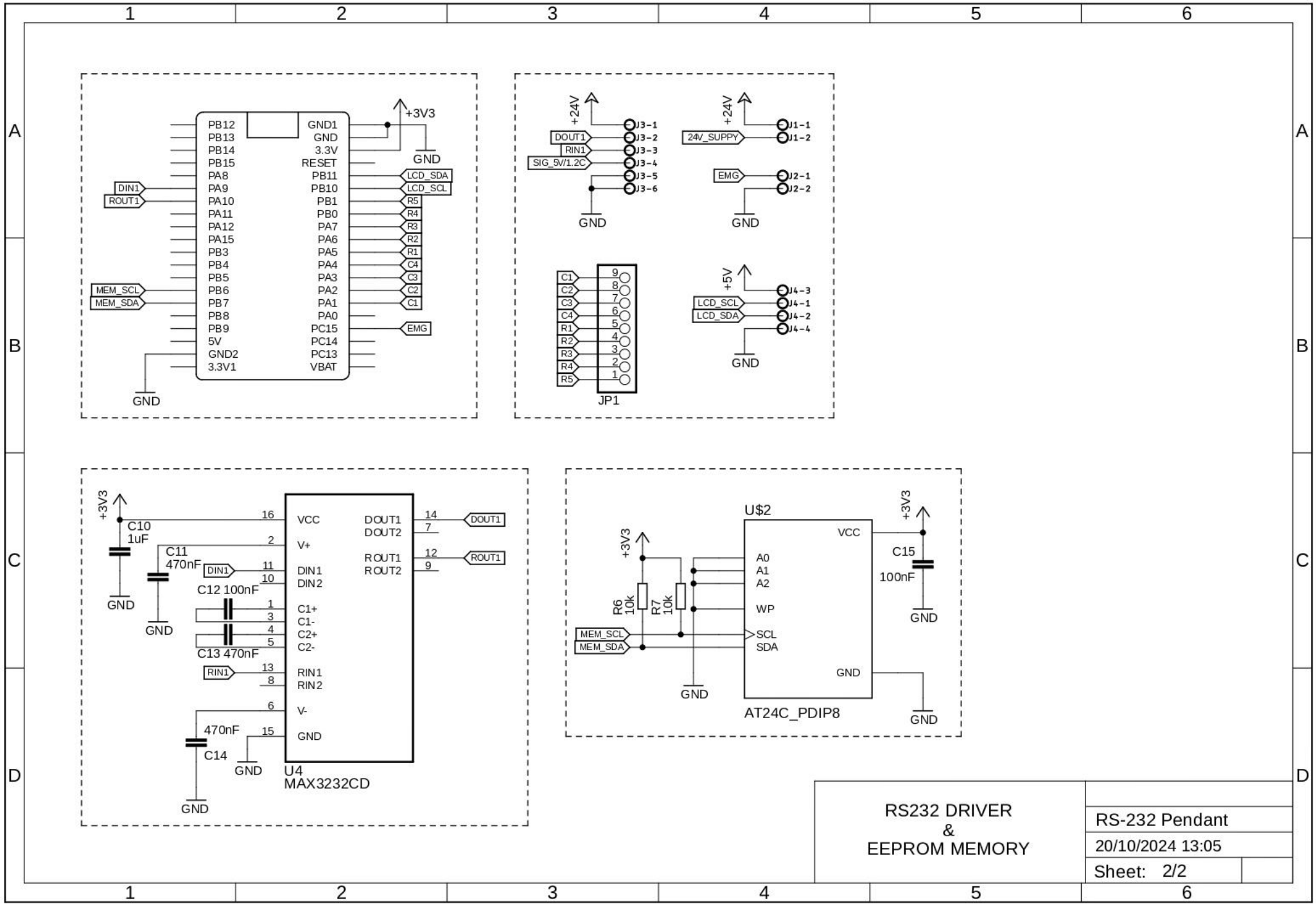
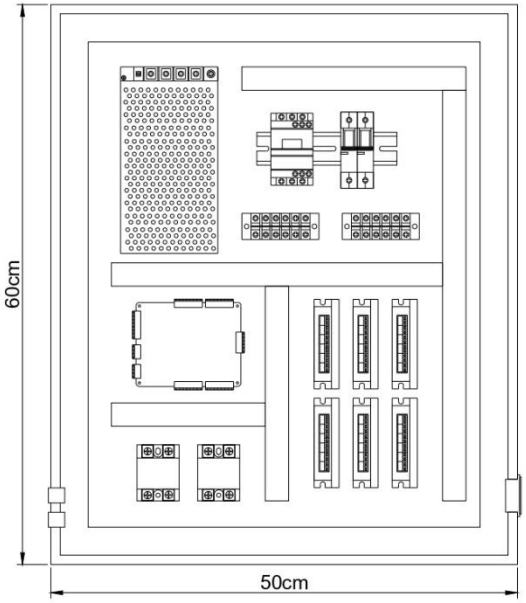


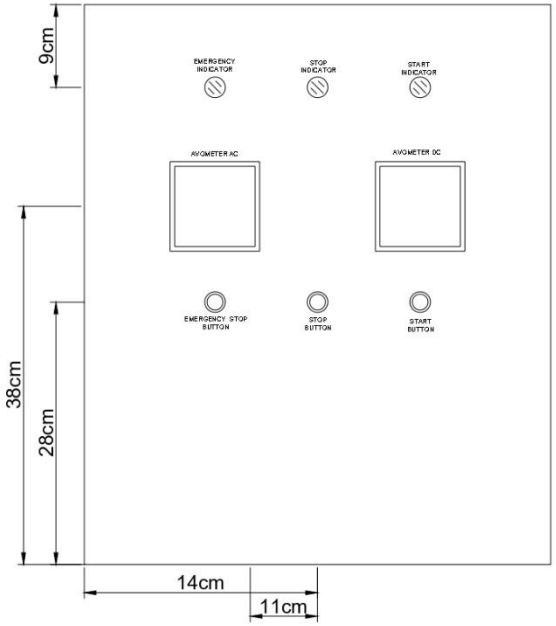
PANEL KONTROL

DESAIN PANEL KONTROL

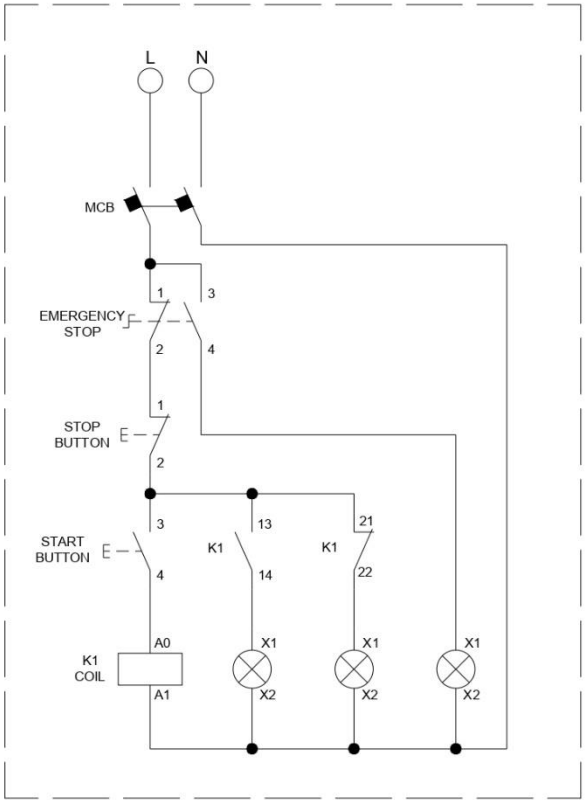
BOX PANEL



TUTUP BOX PANEL



SINGLE LINE DAYA PANEL KONTROL



KOMPONEN DAYA			
No	Nama	Keterangan	Jumlah
1	MCB AC 2 pole	6 Ampere	1
2	Terminal Block	6 Pin 15A	2
3	Pilot Lamp	220V / 22mm	3
4	Emergency Stop	1NO 1NC	1
5	Push Button	1NO	2
6	Avometer AC	500V 20A maks	1
7	Avometer DC	100V 50A maks	1

KOMPONEN KENDALI			
No	Nama	Keterangan	Jumlah
1	PCB Utama	-	
2	Stepper Driver	TB6600	6
3	Solid State Relay	DC To DC 40A	2

LAIN - LAIN			
No	Nama	Keterangan	Jumlah
1	Cable Dust	25x25mm 1,7m	5
2	Terminal Block	220V 10A 6 pole	2

DIGAMBAR	: ALFONSUS GIOVANNI MAHENDRA P.	NIM	: 40040621650042
APLIKASI	: AUTOCAD 2022	TANGGAL	: 15 Oktober 2024

PERANCANGAN BOX PANEL KONTROL
UNTUK WELDING ARM MANIPULATOR 6 DOF

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SEMARANG

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